

Electric field of Point Particles

Questions, Textbook; Answers, Elias Xu

February 3, 2025

4

Two charged particles are attached to an x axis: Particle 1 of charge $-2.00 \times 10^{-7} C$ is at position 6.00 cm and particle 2 of charge $2 \times 10^{-7} C$ is at position $x = 21.0 \text{ cm}$. Midway between the particles, what is their net electric field in unit-vector notation.

Given the fact that the particle in midway, the electrical fields point in the same direction. The resulting calculation is simple:

$$E_{\text{net}} = E_1(r) - E_2(r) =$$

7

In the figure below, the four particles form a square of edge length $a = 5.00 \text{ cm}$ and have charges $q_1 = +10 \text{ nC}$; $q_2 = -20 \text{ nC}$; $q_3 = -20 \text{ nC}$; $q_4 = -10 \text{ nC}$. In unit-vector notation, what net electric field do the particles produce at the square's center?

